Mohammad Abid Hasan

Graduate Research Assistant, Texas Tech University (TTU)

has24395@ttu.edu – (512) 667 3468 – Portfolio

Education

Texas Tech University, Lubbock, Texas, United States *Jan 2025 – Present*
Master of Science in Atmospheric Science
Relevant Coursework: Wind Science and Modeling, Radar Meteorology, Geophysical Fluid Dynamics, Synoptic and Mesoscale Dynamics, Boundary Layer Meteorology
CGPA: 4.00/4.00

Texas State University, San Marcos, Texas, United States *Aug 2024 – Dec 2024*
Master of Science in Civil Engineering
Relevant Coursework: Advanced Reinforced Concrete, Advanced Infrastructure Materials, Probability, Random Variables, & Stochastic Processes for Engineers
CGPA: 4.00/4.00

Bangladesh University of Engineering and Technology, Dhaka, Bangladesh *Mar 2018 – May 2023*
Bachelor of Science in Civil Engineering
Relevant Coursework: Fluid Mechanics, Hydrology, Irrigation and Flood Management, Solid Mechanics, Structural Dynamics
CGPA: 3.52/4.00

Research Interests

- Wind engineering, Building relevant winds (damaging straight-line winds, thunderstorms, tornadoes)
- Disaster-resilient built environment
- AI-accelerated computational fluid dynamics (CFD)
- Urban canopy layer dynamics, urban heat island
- Structural health monitoring, non-destructive evaluation

Publications

Peer-reviewed journal

- **Mohammad Abid Hasan**, Faiaz Khaled, Franklin T. Lombardo, “Advancing tornado-like vortex (TLV) generation in straight-line wind simulators: An LES study,” *Journal of Wind Engineering and Industrial Aerodynamics*, Volume 271, 2026, 106373, ISSN 0167-6105. DOI: 10.1016/j.jweia.2026.106373

Peer-reviewed journal (to be submitted)

- Faiaz Khaled, **Mohammad Abid Hasan**, Franklin T. Lombardo, “Tornado-like vortex (TLV) induced loading in straight-line wind simulators."

Conference abstracts and papers

- Oral presentation at the 2026 AMS Annual Meeting, Houston, TX, on *Generating tornado-like vortices (TLVs) in straight-line simulators using moving louvers with LES*

Peer-reviewed conference abstracts and papers

- **Mohammad Abid Hasan**, Faiaz Khaled, Franklin T. Lombardo (2026), "Advancing Tornado-Like Vortices (TLVs) Modeling in Straight-Line Simulators," *The 8th International Symposium on Computational Wind Engineering*, London, ON, Canada
- **Mohammad Abid Hasan**, Faiaz Khaled, Samuel Hanson, Jon Schroeder (2026), "Digital twin of a supercell rear-flank downdraft (RFD) current in a computational domain using LES," *The 8th International Symposium on Computational Wind Engineering*, London, ON, Canada
- Samuel Hanson, **Mohammad Abid Hasan**, Faiaz Khaled (2026), "Investigating interference effects on aerodynamic loading under tornado-like vortices: An LES study," *The 8th International Symposium on Computational Wind Engineering*, London, ON, Canada

Research Experience

Graduate Research Assistant, Department of Geoscience, Texas Tech University *Jan 2025 – Present*

- Developed large eddy simulation (LES) frameworks to study tornado-like vortex generation in a straight-line wind tunnel
- Investigated interactions between transient wind fields and bluff bodies using LES
- Constructed a "digital twin" of a supercell rear-flank downdraft (RFD) flow structure using LES
- Integrated Weather Research and Forecasting (WRF), FastEddy, and building-scale CFD models to develop a multiscale atmospheric–urban coupling framework

Graduate Research Assistant, Ingram School of Engineering, Texas State University *Sep 2024 – Dec 2024*

- Conducted finite element analyses (FEA) in ANSYS to evaluate and optimize continuously reinforced concrete pavement (CRCP) designs for improved structural performance and durability
- Coordinated field instrumentation and data acquisition campaigns and assessed pavement structural behavior under environmental and service loads

Undergraduate Research Thesis, BUET, Dhaka, Bangladesh *May 2022 – May 2023*

Thesis title: Detecting Damage in Bridges Using Vehicle-Induced Dynamic Response: A Time Series Analysis and ARIMA Modeling Approach

- Developed a damage detection model using finite element method (FEM), time series analysis, and ARIMA modeling
- Identified damage locations and severity in simulated bridge models
- Finite Element Modeling, ARIMA, statistical analysis, structural health monitoring

Teaching Experience

Mentoring Undergraduate Research on Machine Learning for Accelerating Numerical Simulations *Spring 2026*

- Mentored Oluwatimilehin Sapara, Undergraduate Student in Computer Science, Texas Tech University, on the development of an XGBoost surrogate model to accelerate numerical simulations

Lecturer, Department of Civil Engineering, Dhaka International University *Aug 2023 – Jul 2024*

- Courses taught: Engineering Mechanics (0732-1101), Fluid Mechanics Sessional (0732-2262), Computer Programming Sessional (0613-2202), Design of Concrete Structure II (0732-3213)
- Supervised student projects and guided thesis research

Project Experience

Real-Time Structural Health Monitoring for Kyle Fire Station *Sep 2024 – Dec 2024*

- Implemented a self-sensing concrete system with vibrating wire strain gauges
- Analyzed real-time data for infrastructure performance using Loggernet

Undergraduate Capstone Project: Development of a Swimming Pool Complex, BUET *May 2022 – May 2023*

- Designed and analyzed an Olympic-sized pool using ETABS, AutoCAD, and SketchUp
- Conducted feasibility analysis considering environmental, geological, and economic factors
- Estimated costs and prepared a comprehensive Bill of Quantity (BOQ)

Field Campaign

Obtaining "full-scale" pressure, velocity, and loading data from the field at high spatial and temporal resolution for ABL in Lubbock, Texas *25 Jun 2025 – 27 Jun 2025*

- Designed and deployed the StickNet array geometry around an instrumented cube to resolve upstream inflow and wake-region flow
- Collected high-resolution spatial and temporal wind velocity and pressure data around the instrumented cube

"Full-scale" pressure, velocity, and loading data from the field at high spatial and temporal resolution for a vortex-based flow system (dust devil) in Las Cruces, New Mexico *29 Jun 2025 – 3 Jul 2025*

- Deployed TTU StickNets and GoPro cameras to visualize and quantify loading on an instrumented cube induced by a dust devil
- Collected near-ground, high-resolution spatial and temporal thunderstorm outflow data

TxDOT 0-7206: Develop Optimum 2-Mat Reinforcement Design in Continuously Reinforced Concrete Pavement (CRCP) in Sweetwater, Texas *1 Dec 2024 – 5 Dec 2024*

- Designed sensor layout and developed a data acquisition system for field monitoring of CRCP
- Instrumented pavement specimens with vibrating-wire strain gauges (VWSG), electrical resistance strain gauges, and crack meters
- Collected and analyzed field data using dataloggers

Honors & Awards

- General Geosciences Scholarship, Texas Tech University (Fall 2025, Spring 2026)
- Travel grant, Texas Tech University
- Graduate Merit Fellowship, Texas State University (Fall 2024)
- Technical Scholarship, Bangladesh University of Engineering and Technology (2018–2023)
- Talent Pool Scholarship (awarded by the Education Board of Bangladesh)

Skills

- **Programming & Computing:** MATLAB, Python, C++, Linux, parallel computing, High-Performance Computing (HPC)
- **CFD & Simulation:** OpenFOAM, ANSYS Mechanical, ABAQUS
- **Structural & Civil Engineering:** ETABS, SAP2000, AutoCAD
- **GIS & Geospatial Analysis:** ArcGIS
- **Instrumentation & Data Acquisition:** Anemometers (StickNets), Campbell Scientific CR1000X dataloggers, data acquisition systems (DAQ), Arduino Mega microcontrollers, vibrating-wire strain gauges, and electrical resistance strain gauges
- **Machine Learning:** Neural networks (U-Net), regression modeling, surrogate model development
- **Communication:** Technical writing, scientific presentations, public speaking

Extracurricular Activities

- **Member:** Student Chapter of the American Meteorological Society (AMS) at Texas Tech University *Oct 2025 – Present*
- **Member:** South Asian Alliance for Disaster Research Institutes (SAADRI) *Nov 2024 – Present*